Sentiment Analysis of Financial News for Stock Prediction

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Abstract

Stock prediction is critical to algorithmic trading to minimize telegraphed losses and guide investors toward promising trades. Fundamental analysis of markets is critical to address gaps of knowledge from technical analysis in order to accurately forecast stock prices. In this paper, I present a methodology to translate sentiment analysis of relevant financial tweets into profitable trading decisions. I compare the performance of three sentiment analysis models (VADER, BERT, and FinBERT) on a labeled financial Twitter dataset, evaluating their effectiveness in extracting actionable sentiment. My results show that FinBERT outperformed the other models, achieving the highest F1-score in sentiment classification, while VADER unexpectedly outperformed the base BERT model likely due to its rule-based design being better suited to short, informal text like tweets. I then used the sentiment labels to train a stock price prediction model, which achieved a 90% accuracy rate. However, a significant skew toward neutral sentiment predictions introduced potential bias that may have affected the model's performance. These findings highlight both the promise and the limitations of using sentiment-driven models for real-time market forecasting.

1 Introduction

Algorithmic trading has long been a core driving force behind market movements, relying on quantitative indicators such as historical price data, volume, and patterns of trade. However, these trading decisions from the perspective of technical analysis often fail to account for critical factors in why markets move the way they do. These underlying factors of public sentiment can be addressed through fundamental analysis of macro-economic data or textual reports. Recent black-swan-like events, including COVID and unexpected USA tariffs, have further demonstrated that pure quantitative analysis continues to fall short. Public opinion and media narratives affect modern markets at unprecedented speed and magnitude. In particular, with markets moving hundreds of millions of dollars with singular statements by high profile individuals, financial news through platforms like X (formerly known Twitter) have become increasingly influential. But fundamental analysis poses its own set of challenges, especially with unstructured, variable, and dynamic data.

This paper intends to address how to quantify fundamental analysis of a company's performance by corresponding a company's image online to their stock performance. I will accomplish this by evaluating sentiment analysis model performance on financial discussion through X to contribute to a stock price prediction algorithm, informing traders on possible trends indicated in the market. By transforming textual sentiment into structured signals, I explore whether predictive patterns emerge that can be leveraged for short-term trading decisions.

The models I evaluate include the dictionary-based tool VADER and several versions of the pretrained transformer architecture, RoBERTa. Each model is assessed for its ability to classify sentiment in financial contexts and its utility in supporting a trading strategy based on predicted polarity. The broader aim is to quantify whether there exists a reliable relationship between textual sentiment and subsequent price movement, and, if so, how that relationship can be modeled to support automated trading systems.

2 Related Work

Stock trend prediction has a decisive role in stock investment, allowing investors to make better decisions on whether to buy, sell, or hold according to the most likely-to-occur future markets. Thus, stock trend prediction has attracted many research efforts, traditionally categorized into technical and fundamental analysis. These forms of analysis are based on different data. Technical analysis studies chart patterns, price movements, and volume while fundamental analysis studies a company's financial health, earnings, and other indirect factors to identify potential trading opportunities [9]. The main goal of this type of approach is to discover the trading patterns that can be leveraged for future prediction.

Sentiment analysis has become increasingly critical to integrate qualitative public perception into quantitative trading strategies based on future forecasting. Early work establishes with psychology that investor sentiment foreshadows price fluctuations and trading volumes [11].

2.1 Dictionary-based Models

Initial approaches to financial sentiment analysis relied on lexicon-driven methods, which assign sentiment scores to text based on curated lists of positive and negative words [6]. Fesenko noted that these models, while lightweight, often underperform in domains like finance where language is complex and context-dependent [7]. A broad review by Nassirtoussi et al. [5] highlights their limitations when faced with implicit sentiment, sarcasm, or specialized terminology. Tools like VADER [14], originally designed for social media content, have been adapted for financial applications due to their ability to handle informal writing, punctuation, and capitalization, addressing lexicon-based challenges with limited success.

Thus, sentiment analysis shifted to more sophisticated representations using machine learning and transformer architecture to better capture contextual relationships. Neural architectures like convolutional neural networks (CNNs) and long short-term memory networks (LSTMs) learned hierarchical structures from data and improved performance in tasks requiring semantic understanding. Chahuán-Jiménez [1] demonstrated the utility of such models in forecasting stock index trends, capturing nonlinear interactions between textual sentiment and market behavior more effectively than static models.

2.2 Natural Language Processing

More recently, transformer-based models like BERT [4] have revolutionized natural language understanding by modeling bidirectional context in text, as discussed in the Hugging Face LLM course:

> *The Transformer architecture was introduced in June 2017. The focus of the original research was on translation tasks. This was followed by the introduction of several influential models, including:*
> *June 2018: GPT, the first pretrained Transformer model, used for fine-tuning on various NLP tasks and obtained state-of-the-art results.*
> *October 2018: BERT, another large pretrained model, this one designed to produce better summaries of sentences.*

These pre-trained models, when fine-tuned on financial datasets, outperform previous methods in sentiment classification tasks. FinBERT, a variant fine-tuned on financial texts, shows strong results in sentiment analysis accuracy with both casual and financial text [3].

An important development is the use of heterogeneous architectures that incorporate both text and structured financial data. Hu et al. [2] proposed a framework imitating human neural processing that models news-based sentiment with historical price data to predict market trends, reflecting a broader movement toward multimodal deep learning models.

3 Assumptions

Sentiment classifiers in financial contexts rely on a range of theoretical and practical assumptions that simplify real-world complexities of financial markets. Understanding these assumptions is critical to contextualizing my model performance, generalizability, and the potential for bias or overfitting.

3.1 Semantic Consistency

I assume that semantic relationships trained from static and general datasets apply consistently to financial tweets. This is not necessarily always the case, as general and domain-specific language shifts over time, resulting in non-stationary relationships between financial sentiment and markets. Although I use domain-adapted transformer models with further fine-tuning on a tweet-specific dataset, this does not guarantee the viability of the model's performance on text outside of the static context the model is trained and tested on.

3.2 Dataset Independence and Representativeness

Many financial tweets are ambiguous as to which category they belong in, which introduce label noise and class imbalance. Despite this, I assume that the sentiment labels on the Twitter Financial News Sentiment Dataset are accurate, consistent, and represent true investor sentiment.

In the real-world, general financial sentiment is dependent on contextual knowledge. For the sake of mathematical modeling, I assume each text input (tweet, headline, etc.) is independent and uniformly distributed.

Certain rare events (e.g., pandemics, geopolitical shocks) and new companies or instruments fall outside the training distribution. However, in order to generalize myresults beyond 2020, I assume that semantic consistency will hold so the training data reflects the full diversity of future inputs and market events.

3.3 Rational Market Reaction

Investor behavior can be driven by external constraints (e.g., regulation), which weaken sentiment's predictive power. Moreover, aligning the timing of news releases with stock price reactions introduces uncertainty, as markets may respond with delay, overreaction, or not at all depending on context and existing volatility. For stock price prediction, I assume that markets correspond to sentiment efficiently within a day's time.

4 Mathematical Models

4.1 Natural Language Toolkit Valence Aware Dictionary and sEntiment Reasoner (NLTK VADER)

NLRK VADER is a lexicon and grammatical/syntactical rule-based sentiment analysis tool trained specifically on social media sentiment, including emojis [14]. A lexicon-based model uses a predefined dictionary of words, phrases, acronyms, or emojis associated with sentiment scores rated for polarity and intensity. For VADER, the sentiment of a given document or sentence is computed as a function of the sentiment polarity of the words it contains and its structure, since structure can modify polarity and intensity. This calculation is given by the sums of scores from features found in the text and normalized:

$$compound\ score\ = \frac{x}{\sqrt{x^2 + \alpha}}$$

where alpha is set to 15, the approximate maximum to the expected value of x [15]. Due to its simplicity, VADER is exceptionally fast computationally.

4.2 Bidirectional Encoder Representations from Transformers (BERT)

Transformer models are an alternative to Recurrent Encoder-Decoder architectures that solve the problem of modeling sequences of variable length. BERT is built by a stack of Transformer encoder layers using masked language modeling (MLM) and next sentence prediction (NSP) [14].

Input tokens are transformed into a vector embedding summing token, segment, and position embeddings. This input is passed through 12 Transformer Encoder Layers. Encoder and decoder structures work by encoding the source sequence into a new representation and decoding the representation into another sequence as close to the target sequence as possible [17].

BERT follows the same encoder-decoder structure, but doesn't incorporate recurrency. Rather, it builds an attention mechanism that doesn't depend on sequential processing of any recurrent unit ie. leveraging self-attention to model dependencies across all words in a sentence simultaneously. It produces contextual embeddings for each token and can be fine-tuned for classification tasks [16].

Each layer of BERT applies multi-head attention, where attention between tokens is computed as:

$$Attention(Q, K, V)\ =\ softmax(QK^T / \sqrt{d_k})V$$

where $Q, K, V\ \in\ \mathbb{R}^{n \times d_k}$ are query, key, and value matrices from input embeddings, n is the sequence length, and d_k is the attention dimension [18]. The multi-head self-attention module propagates information from the encoder to the decoder.

After each encoder and decoder sublayer is *Add&Norm*, which performs residual learning and layer normalization on multi-head attention and feedforward modules:

$$Add\&Norm\ =\ LayerNorm(X\ +\ (Attention\ or\ FFN))$$

The output of the final encoder layer is a sequence of vectors one-to-one with input tokens. This is then followed by MLM, which randomly masking tokens and training the model to predict them, and NSP, which is predicting sequential segments.

4.3 Fine-tuned BERT

To post train BERT for financial tweets, I adjust all weights with backpropagation and cross-entropy loss using data:

$$D\ =\ \{(x_i, y_i)\}_{i=1}^{N}$$

where x_i is the tokenized input text, y_i is the sentiment class, and N is the number of examples. Let θ be the pretrained parameters of BERT and ϕ be the task-specific classifier head. Each token x_i is passed through BERT with parameters θ into the output h_i. This is passed to a linear classifier to learn parameters W and b :

$$y_i = softmax(Wh_i\ +\ b)$$

Then, θ and ϕ are updated using gradient-based Adam optimizer to minimize cross-entropy loss. This optimizes BERT to the financial tweet dataset.

5 Implementation[1]

5.1 Datasets

I draw on three datasets for sentiment training, classification, and correlation to stock price. The ZeroShot Twitter Financial News Sentiment Dataset is a comprehensive dataset relating tweets including stock tickers with sentiment, including bearish, bullish, and neutral [8]. The Tweets About the Top Companies from 2015 to 2020 dataset relates tweets with timestamps from 6 major

[1] The full code is available at: Colab Notebook.

companies to stock tickers. This dataset is mapped to YFinance closing prices in order to relate sentiment to stock price fluctuations.

All datasets were preprocessed by standardizing dates and congregating duplicate tweets into sums of retweets. Due to computational limitations, I only evaluate model performance in predicting Microsoft stock.

5.2 Sentiment Analysis

The implementation of my sentiment analysis pipeline involved three primary approaches: VADER, BERT, and a FinBERT model fine-tuned specifically for financial text.

For the rule-based approach, I used VADER (Valence Aware Dictionary and sEntiment Reasoner), a sentiment analysis tool tuned for social media text. I loaded the "twitter-financial-news-sentiment" dataset from Hugging Face, which provides labeled tweets related to finance. Using the VADER sentiment analyzer, I computed compound scores for each tweet and mapped these scores to one of three sentiment labels—positive, negative, or neutral—based on standard VADER thresholds. I then compared the predicted labels to the true labels to evaluate model performance using accuracy and a detailed classification report.

Next, I implemented a pretrained BERT model, specifically the `twitter-roberta-base-sentiment` model developed by Cardiff NLP, which is trained on general Twitter sentiment data. I tokenized each tweet and passed it through the model to obtain class logits, which I then converted into predicted sentiment labels using a softmax function. I repeated the evaluation process using standard classification metrics.

To improve domain-specific accuracy, I employed the FinBERT model, which is fine-tuned on financial Twitter data. Using the same methodology of tokenization, inference, and evaluation, I found that FinBERT generally outperformed the other two models in financial sentiment classification.

To compare model performance more thoroughly, I extracted precision, recall, and F1-scores for each sentiment class from the classification reports and visualized them using seaborn's categorical bar plots.

5.3 Stock Price Prediction

After identifying fine-tuned FinBERT as the best model for sentiment analysis of financial tweets, I processed a real-world dataset containing tweets about top companies between 2015 and 2020, obtained from Kaggle. I merged company identifiers with tweet text data and I analyzed the distribution of tweet frequency across companies. I selected Microsoft (ticker: MSFT) to study and filtered tweets relevant to the company, and then ran the fine-tuned FinBERT to create a sentiment column.

Following generating individual tweet sentiments, I gather closing price data from YFinance and map it to tweets by grouping tweets by day to create weighted retweet- and like-sentiment metrics associated with closing price. These features are incorporated into a RandomForestClassifier model to predict bearish, bullish, and neutral price movements. This is compared to the next-day percent change of price, where a magnitude of price change greater than 1 percent was categorized as Bullish or Bearish correspondingly, to verify the accuracy of the model predictor.

6 Results and Analysis

6.1 Sentiment Model Performance

FinBERT post-trained on the Twitter Financial News Sentiment Dataset performed the best out of all three models, achieving an 86.5% accuracy (Figure 1). This conclusion is sensible since FinBERT is already tuned to financial contexts. With further optimizations in tuning FinBERT to the dataset, the model would better encapsulate the sentiment behavior for financial tweets, which generally have a specific structure unique from long-form text that BERT is trained on.

VADER outperformed the base BERT model in my analysis with a 49.6% accuracy compared to 36.8%, which may be attributed to several factors. First, VADER is a rule-based model specifically designed for social media and short-form text, making it particularly well-suited to the informal and concise language of financial tweets. It effectively captures sentiment cues such as capitalization, punctuation, and emoticons, which BERT may overlook or misinterpret.

Additionally, BERT was trained on general-purpose corpora like Wikipedia and BookCorpus, which do not reflect the informal, abbreviated, and context-specific language found on platforms like Twitter. Without additional fine-tuning on financial or social media text, BERT's contextual embeddings may struggle to accurately interpret the sentiment of brief, jargon-heavy financial posts. This highlights that even powerful language models like BERT can underperform compared to simpler methods when not adapted to the specific linguistic features of the target domain.

6.2 Stock Price Prediction Evaluation

The stock price prediction model achieved a notable accuracy rate of 90%, suggesting that the sentiment analysis provided a strong signal for market movement prediction. However, this high accuracy may be somewhat misleading due to an overwhelming bias in the sentiment classification toward the "neutral" label.

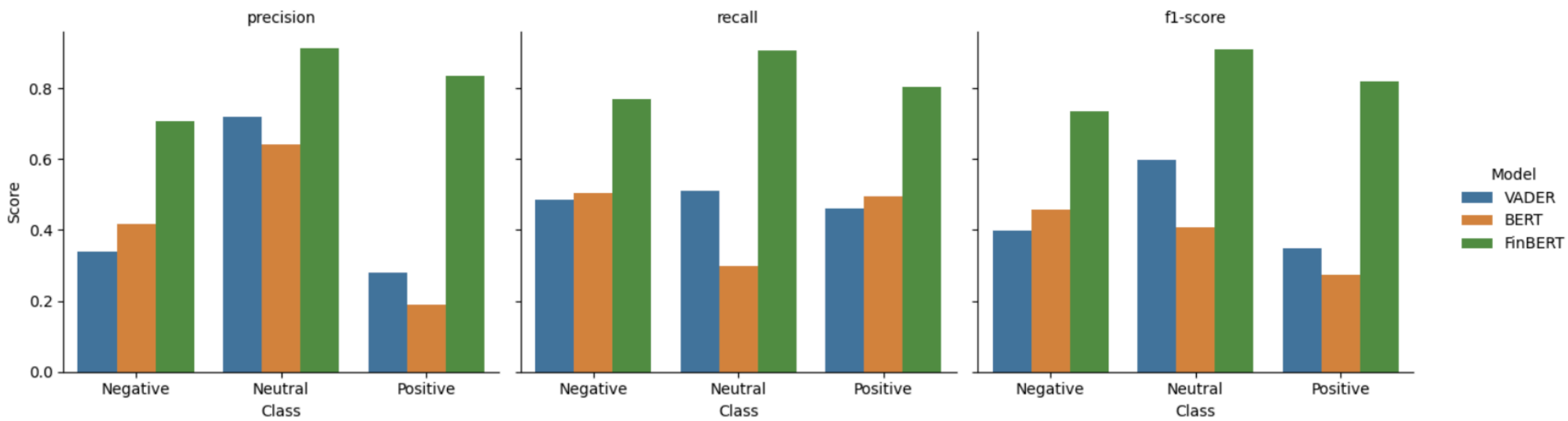

Figure 1. Classification scores of VADER, BERT, and Fine-Tuned FinBERT models.

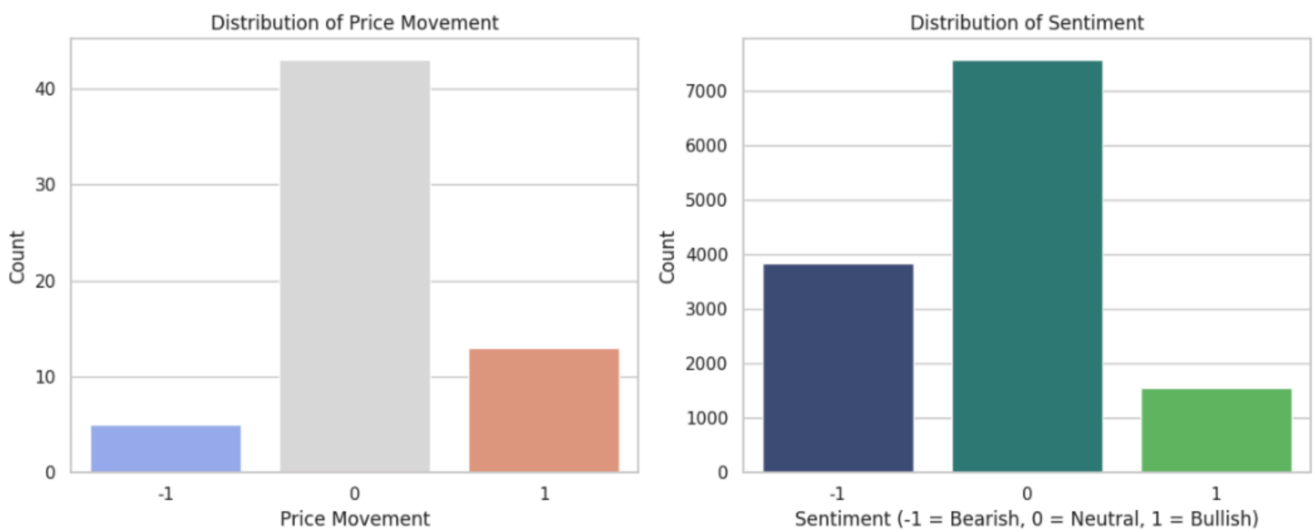

Figure 2. Distributions of Bearish, Neutral, and Bullish labels in MSFT stock price (left) and MSFT tweet sentiment (right).

Both BERT and FinBERT exhibited a tendency to categorize a large portion of tweets as neutral, which may have skewed the training data and influenced the prediction outcomes. This imbalance in sentiment labeling could result in a model that performs well statistically but lacks true predictive nuance, especially in capturing the emotional drivers of stock volatility.

Consequently, while the accuracy rate is encouraging, it should be interpreted cautiously in light of the potential bias introduced by dominant neutral classifications.

7 Limitations and Future Work

7.1 Limitations

One limitation of my study lies in the imbalance of sentiment classes within the dataset, which may have impacted model performance particularly for the neutral and negative classes (Figure 2). VADER, being a rule-based model, struggled to accurately capture the nuances and contextual meanings present in financial language, especially in cases involving sarcasm, abbreviations, or domain-specific jargon. Although FinBERT outperformed the other models, its predictions may still reflect biases from the data it was fine-tuned on, and it may not generalize well to out-of-distribution data or newly emerging financial language. Additionally, my evaluation focused primarily on accuracy and F1-score, without analyzing the temporal alignment between sentiment trends and actual stock performance, which is crucial in financial forecasting.

7.2 Future Research Directions

For future research, expanding the dataset to include a broader range of financial entities and more recent tweets could improve model robustness and relevance. Incorporating time-series sentiment analysis in tandem with stock price movements would provide more actionable insights and allow for causal inference between sentiment and market outcomes.

To address assumptions that could invalidate my model, integrating Granger causality tests to explicitly measure the lead-lag relationship between sentiment signals and market behavior could further inform trading and modeling decisions..

Furthermore, experimenting with ensemble methods or large-scale language models like GPT-4, fine-tuned specifically for financial sentiment, may enhance predictive accuracy. Implementing online learning or continual fine-tuning to adapt models over time can help maintain temporal relevance by periodically updating models with new data.

Integrating additional metadata such as user influence, tweet engagement, or market context could enrich model inputs and improve sentiment interpretation in a real-world financial setting.

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